Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **Node.js** and **TypeScript** microservices, containerized with Docker and **Kubernetes**, on SQL and NoSQL databases with analytics, deployed to AWS and Azure for scalable, secure customer-facing systems.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Designed normalized multi-tenant **SQL** schemas with row-level security for 500+ users, keeping each tenant's data isolated and tuning indexed queries to cut dashboard latency from 1.2s to 350ms.
- Built backend services with fault-tolerant job processing and integration pipelines across PostgreSQL and cloud storage, tripling throughput across 1K+ jobs at 99%+ **reliability** through CI/CD releases across 5K+ monthly jobs.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built and deployed a customer-facing platform on **AWS** with React and REST APIs, serving 150+ instructors and 10K+ daily requests with stable responses during peak load through GitHub Actions CI/CD.
- Captured instructor requirements for upload, rubric, and grading workflows, then shipped multimodal AI features with fallbacks and **per-call cost tracking** that cut instructor feedback turnaround 60% in production use.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Built cross-platform **microservices** with object-oriented Factory and Singleton patterns, covered by 20+ automated unit and integration test suites in nightly Jenkins pipelines across Windows, Linux, and macOS.
- Redesigned AWS Glue and S3 pipelines that consolidated activity logs for **analytics**, boosting efficiency 68% and cutting storage costs 15%, and optimized APIs to improve response times 30%.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built 8 **Node.js** and Express microservices in **TypeScript** on MongoDB and Azure SQL for authentication, resource allocation, billing, and notifications, syncing 2M+ records across locations with WebSockets.
- Packaged services as **Docker** images through Jenkins CI/CD to Azure with automated tests and zero-downtime rolling releases, and used product analytics to cut resource wastage 42% and lift lead conversions 15%.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built booking, pricing, and payment APIs with **NoSQL** Firebase real-time sync and time-slot locking to prevent double bookings, keeping responses under 200ms across 50K+ records for mobile and web clients.
- Built hourly Spark analytics on Databricks over booking and payment events, and automated refunds and disputes through idempotent webhooks, cutting manual refund handling about 70% for operations teams.

Skills

Languages: TypeScript, JavaScript, Python, Java, SQL
Backend: Node.js, Express.js, REST APIs, Microservices, WebSockets, Webhooks, FastAPI, Flask
Containers & Cloud: Docker, Kubernetes, AWS (EC2, S3, RDS, Glue), Azure, GCP, CI/CD, Jenkins, GitHub Actions
Data & Analytics: PostgreSQL, MySQL, Azure SQL, MongoDB, Firebase, NoSQL, Schema Design, Spark, Databricks
Practices: Unit Testing, Jest, PyTest, Git, Agile, Debugging, Requirements Gathering, React, Linux, Code Reviews

Projects

EV Charging Session Service | *Node.js, TypeScript, Kubernetes, PostgreSQL, Redis*

- Designed a **Node.js** microservice on Kubernetes that starts and stops charging sessions over the OCPP protocol, tracks charger status in Redis, and records energy and billing data in Postgres.

Charging Network Analytics | *TypeScript, DynamoDB, AWS Lambda, Athena*

- Designed a pipeline that stores charger events in **DynamoDB**, rolls them into hourly utilization and fault reports, and flags stations with repeated failed sessions for **field service**.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB, WebSockets*

- Built a **TypeScript** platform deployed on **Kubernetes** with CI/CD, Plaid bank linking, and real-time WebSocket updates, supporting 50,000+ users and 1M+ transactions with MongoDB and a token-based firewall.